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APPARATUS, SYSTEM, AND METHODS FOR CULTIVATING GREEN MICRO ALGAE (H. PLUVIALIS) AND FOR HARVESTNG ASTAXANTHIN THEREFROM

Abstract

An assembly, methods, and network for farming and culturing green microalgae (*Haematococcus pluvialis*) are disclosed, comprising: a mechanical support framework; a plurality of cultivation pouches, mechanically coupled to the mechanical support framework, having arrays of cultivation pouches where green microalgae (*H. pluvialis*) are cultured and harvested; a plurality of light emitting diodes (LEDs), mechanically coupled and supported by the mechanical support framework and arranged in between two of any adjacent arrays of cultivation pouches, operable to provide a predetermined light intensities to each of the cultivation pouches; and an air and gas supply and distribution unit, mechanically coupled and supported by the mechanical support framework, operable to supply predetermined amounts of air and carbon dioxide (CO.sub.2) to each of the green microalgae pouches.

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Background/Summary

CLAIM OF PRIORITY

[0001] This application claims priorities under 35 U.S.C. § 119 (a)-(d) of Applications No. 1-2024-05237 entitled "Hcustom-character Thcustom-characterng Scustom-charactern Xu custom-charactert Vi Tcustom-charactero Lcustom-characterc *Haematococcus Pluvialis*", filed on Jul. 16, 2024; which claims priority of a patent application No. 1-2024-04312 entitled "Ph custom-charactercustom-characterng Pháp Scustom-charactern Xucustom-charactert Astaxanthin tcustom-character Vi Tcustom-charactero Lcustom-characterc *Haematococcus Pluvialis*" filed on Jun. 12, 2024; which claims priority of a patent application No. 1-2024-01264 entitled "Thicustom-charactert Bcustom-character Nuôi Vi Tcustom-charactero L custom-characterc *Haematococcus Pluvialis*" filed on Feb. 22, 2024; which claims priority of a patent application No. 1-2024-01265, entitled "Phcustom-charactercustom-characterng Pháp Nuôi Vi Tcustom-charactero Lcustom-characterc *Haematococcus Pluvialis*" filed on Feb. 22, 2024; all in the Republic Socialist of Vietnam. These patent applications identified above are incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

[0002] The present invention relates generally to farming apparatus and technique. Specifically, the present invention referring to system and apparatus for farming and harvesting green microalgae *Haematococcus pluvialis* (*H. pluvialis*) and astaxanthin.

BACKGROUND ART

[0003] Among many species of microalga, *Haematococcus pluvialis* has the highest source of natural astaxanthin that they are called "super anti-oxidant." Currently, the majority of astanxathin used in aquaculture is synthetic. Natural astaxanthin produced by *H. pluvialis* has significantly greater antioxidant capacity than the synthetic one. Astaxanthin has been used in the nutraceuticals, cosmetics, food, and aquaculture industries. It is now evident that, astaxanthin can significantly reduce free radicals and oxidative stress that improves human health. In addition, green microalgae (*H. pluvialis*) can act as a phytoremediation agent that absorbs pollutants such as carbon dioxide (CO2) and convert to biomass with high value compounds such as astanxanthin.

[0004] For the above reasons, the need to cultivate green microalgae (*Haematococcus pluvialis 'H.*

pluvialis') on an industrial scale for large yields and high astaxanthin content is ever increased. Technical improvements in the farming process has been applied over the years with the aim of achieving economic efficiency, high yield, stable quality, and industrial scale. Many algae farming models attempting to meet the above needs have been researched and developed.

[0005] One of the major problems in the current large-scale algae farming models is economic efficiency. That is, when culturing algae in large volume, bacterial infections often occur and cannot be avoided and controlled. Consequently, when infections occur, all algae in the same tank must be completely eliminated, leading to a huge economic loss.

[0006] Cultivation of microalgae requires light, carbon dioxide (CO.sub.2), nutrients (nitrogen, phosphorus, and sulfur), and various trace elements. Controlling light to allow algae to grow in the green phase as well as in the inhibition phase requires different light controlling at different stage of cultivation. Using a light source to evenly illuminate the algae cultivation volume is of major concern. The light source must be able to change intensity, time, and location to avoid bacterial contamination of the algae.

[0007] Previous attempts to solve the problems related to the cultivation of green microalga *H. pluvialis* include:

[0008] In the U.S. Pat. No. 6,022,701 entiled "Procedure for Large-Scale Production of Astaxanthin From Haematococcus" by Bousillba et al., issued on Feb. 8, 2000, a *Haematococcus* algael cell culture process for the large-scale production of astaxanthin-rich *Haematococcus* cells includes: culturing *Haematococcus* cells in conditions suitable for optimal vegetative growth under a light intensity of about 30-140 μmol photons.m.sup.−2.S.sup.−1 and at a temperature of about 15° C.-28° C.; and collecting the grown cells that was cultured to accumulate astaxanthin in water supplemented with CO.sub.2 at temperatures below 35° C. However, the astaxanthin content of the disclosed process is only about 4%.

[0009] In another US patent application No. US2015/0252391A1, entitled "Method using Microalgae for High-Efficiency Production of Astaxanthin" by Li et al., published on Sep. 10, 2015, a method of producing astaxanthin or increasing astaxanthin content in microalgae is disclosed which includes: culturing microalgae by heterotrophic method; which adds a heterotrophic medium at pH 4.0-10, at a culture temperature of 10-40° C. and with dissolved oxygen above 0.1%; and collecting and diluting the heterotrophic microalgae culture medium and then adding photo induction medium to the heterotrophic microalgae culture medium for photo induction; or directly generating photochemistry to cultivate microalgae to obtain heterotrophs. The disclosed steps aim to produce astaxanthin or increase astaxanthin content in microalgae, at photo induced temperature of 5° C.-50° C., with continuous or intermittent light intensity of 0.1-150 klx for periods from 1 hour to 480 hours. However, the astaxanthin content of the disclosed method is only less than 3%.

[0010] In the hot climate conditions of Vietnam, the infection from the air of fungal spores during microalgae cultivation processes in the algae farming area is very large. Even if controlled, this infection cannot be avoided.

[0011] Therefore, there is a need for a microalgae farming apparatus capable of controlling light intensities at any location and any phase of microalgae cultivation.

[0012] There is a need for a large-scale microalgae farming apparatus and process that are simple, convenience, and cost saving.

[0013] There is a need for a microalgae farming apparatus and process that produces high yield of astaxanthin.

[0014] There is a need for a microalgae farming apparatus and process that minimizes viral infection from the air.

[0015] The methods and apparatus of the present invention meet the above needs and solve the above-described problems.

SUMMARY OF THE INVENTION

[0016] Accordingly, an object of the present invention is to provide Assembly, methods, and network for farming and culturing green microalgae (*Haematococcus pluvialis*) are disclosed, comprising: a mechanical support framework; a plurality of cultivation pouches, mechanically coupled to the mechanical support framework, having arrays of cultivation pouches where green microalgae (*H. pluvialis*) are cultured and harvested; a plurality of light emitting diodes (LEDs), mechanically coupled and supported by the mechanical support framework and arranged in between two of any adjacent arrays of cultivation pouches, operable to provide a predetermined light intensities to each of the cultivation pouches; and an air and gas supply and distribution unit, mechanically coupled and supported by the mechanical support framework, operable to supply predetermined amounts of air and carbon dioxide (CO.sub.2) to each of the green microalgae pouches.
[0017] Another object of the present invention is provide a method of cultivating and harvesting green microalgae (*H. pluvialis*) that comprises: (a) constructing a mechanical support framework that supports habitats for green microalgae at predetermined light intensities and predetermined pressures and temperatures; (b) forming the habitats for the green microalgae (*H. pluvialis*) by using an optimal biomass growing media; (c) proliferating the green microalgae (*H. pluvialis*) by controlling a plurality of LED lights and a CO.sub.2 source and an air pump to generate a predetermined light intensities at 10 μmol photons/m.sup.2/s for 5 days and a temperature of 20° C. to 25° C. and; and (d) proliferating the microalgae (*H. pluvialis*) further by varying the temperatures from 28° C. to 30° C. and the predetermined light intensities to 400 μmol photons/m.sup.2/s.
[0018] Another object of the present invention is to obtain a method and apparatus for farming and harvesting of green microalgae that have high astaxanthin yield above 5%.
[0019] Another object of the present invention is to obtain a method and apparatus for farming and harvesting of green microalgae that minimizes viral infection during the all cultivation phases.
[0020] Another object of the present invention is to obtain a method and apparatus for farming and harvesting of green microalgae that achieves industrial scale production.
[0021] Another object of the present invention is to obtain a method and apparatus for networking a plurality of microalgae apparatuses together that can share information to reduce infection, errors, and to increase overall efficiency and total production.
[0022] These and other advantages of the present invention will no doubt become obvious to those of ordinary skill in the art after having read the following detailed description of the preferred embodiments, which are illustrated in the various drawing and figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0023] The accompanying drawings, which are incorporated in and form a part of this specification, illustrate embodiments of the invention and, together with the description, explain the principles of the invention.
[0024] FIG. **1** shows an overall conceptual representation of a green microalgae (*Haematococcus pluvialis*) farming assembly (apparatus) in accordance with an exemplary embodiment of the present invention.
[0025] FIG. **2** shows a perspective view of the mechanical support framework of the green microalga (*H. pluvialis*) farming assembly in accordance with an exemplary embodiment of the present invention.
[0026] FIG. **3** shows a perspective arrangement of the cultivation pouches (chambers) of the green microalga farming assembly in accordance with an exemplary embodiment of the present invention.

[0027] FIG. **4** shows the arrangement of the LED lights and electrical wiring unit of the green microalga farming assembly in accordance with an exemplary embodiment of the present invention.
[0028] FIG. **5** shows the air and gas supply and distribution unit of the green microalga farming assembly in accordance with an exemplary embodiment of the present invention.
[0029] FIG. **6** shows a 2D perspective diagram of a cultivation pouch in accordance with an exemplary embodiment of the present invention.
[0030] FIG. **7** shows a flow chart of a method for constructing a green-microalgae (*H. pluvialis*) farming assembly that provides high astaxanthin yield in accordance with an exemplary aspect of the present invention.
[0031] FIG. **8** shows a flow chart of a process for growing green microalgae with high astaxanthin yield in accordance with an exemplary aspect of the present invention.
[0032] FIG. **9** shows a flow chart of a software application program for monitoring and controlling the cultivation of green microalgae with high astaxanthin yield in a network of farming assemblies in accordance with an exemplary aspect of the present invention.
[0033] FIG. **10**A-FIG. **10**C show the morphology of the green microalgae cells at different stages —proliferation stage, late proliferation stage, and accumulation stage—under 100× magnification of an electron microscope.
[0034] FIG. **11** presents the HPLC-High Performance Liquid Chromatography (HPLC-High Performance Liquid Chromatography) chart of green microalgae (*H. pluvialis*) powder with the astaxanthin content of 5.09% raised from the green microalgae farming assembly of the present invention.
[0035] FIG. **12** shows the HPLC high-performance liquid chromatography chart of green microalgae (*H. pluvialis*) powder with the astaxanthin content of 7.92% raised from the green microalgae farming assembly of the present invention.
[0036] FIG. **13** shows the HPLC high-performance liquid chromatography chart of an extract sample from green microalgae (*H. pluvialis*) powder with the astaxanthin content of 10.9% raised from the green microalgae farming assembly of the present invention.
[0037] FIG. **14** shows a system level schematic diagram of a green microalgae farming network that connects green microalgae (*H. pluvialis*) farming assemblies together in accordance with an exemplary embodiment of the present invention.
[0038] FIG. **15** shows a schematic diagram of one of a server computer loaded with a software application for monitoring and controlling the network of the green microalgae (*H. pluvialis*) farming assemblies in accordance with an exemplary embodiment of the present invention.
[0039] FIG. **16** shows a graphic user interface (GUI) of a monitor and control software application for the network of the green microalgae (*H. pluvialis*) culture assemblies in accordance with an exemplary embodiment of the present invention.
[0040] The above figures are for the purposes of illustration only. A person of ordinary skill in the art will readily recognize from the following discussion that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles of the technology described herein.

DETAILED DESCRIPTION OF THE INVENTION

[0041] Reference will now be made in detail to the preferred embodiments of the invention, examples of which are illustrated in the accompanying drawings. While the invention will be described in conjunction with the preferred embodiments, it will be understood that they are not intended to limit the invention to these embodiments. On the contrary, the invention is intended to cover alternatives, modifications and equivalents, which may be included within the spirit and scope of the invention as defined by the appended claims. Furthermore, in the following detailed description of the present invention, numerous specific details are set forth to provide a thorough understanding of the present invention. However, it will be obvious to one of ordinary skill in the art that the present invention may be practiced without these specific details. In other instances, well-known methods, procedures, components, and circuits have not been described in detail so as not to unnecessarily obscure aspects of the present invention.

[0042] Within the scope of the present description, the reference to “an embodiment” or “the embodiment” or “some embodiments” means that a particular feature, structure, or element described with reference to an embodiment is comprised in at least one embodiment of the described object. The sentences “in an embodiment,” “in the embodiment,” or “in some embodiments” in the description do not, therefore, necessarily refer to the same embodiment or embodiments. The features, structures, or elements can be furthermore combined in any adequate way in one or more embodiments.

[0043] Beginning with FIG. **1**, an overall conceptual design of a green microalgae (*Haematococcus pluvialis*) farming assembly or apparatus **100** (hereinafter referred to as “farming assembly **100**”) in accordance with an exemplary embodiment of the present invention is illustrated. Farming assembly **100** is a modular, portable, high-yield, and infection free device that is inexpensive to build and easy to assemble. Farming assembly **100** includes five main units that work together to achieve the above-listed objectives of the present invention: a mechanical support framework **200** (framework **200**), an array of green microalgae cultivation chambers or pouches **310** (array of cultivation pouches **310**), an array of light emitting diode (LEDs) and electrical wiring unit **410** (array of LEDs **410**), an air and gas supply and distribution unit **510** (air and gas supply unit **510**), and a monitoring and controlling unit **520** (computer **520**). Mechanical support framework **200** is designed to house, support, and protect array of cultivation pouches **310**, array of LEDs unit **410**, air and gas supply unit **510**, and computer **520**. Array of cultivation pouches **310** contains a substrate (medium) where green microalgae (*H. pluvialis*) proliferates and accumulates until harvest. Array of LEDs **410** is arranged next to each of array of cultivation pouches **310** to provide vital light sources to the green microalgae. Air and gas distribution unit **510** provides air and CO.sub.2 gas to array of cultivation pouches **310**. As its name suggests, computer **520** receives feedback information regarding the substrate inside array of cultivation pouches **310** and issues appropriate instructions. These instructions adjust the correct amount of light, air, temperature, pressure, and CO.sub.2 gas to array of cultivation pouches **310**. In other embodiments of the present invention, computer **520** is networked with farming assemblies **100** from different geographical locations. This networking feature of the present invention helps realizing the modularity feature, sharing vital information that reduce errors, avoiding infectiosn, increasing efficiency and thus industrial production scale of each farming assembly **100**. Farming assembly **100** also provides means to realize different methods for culturing and harvesting green microalgae (*H. pluvialis*) of the present invention.

[0044] Next, FIG. **2** to FIG. **5**, individually or in combination, disclose exemplary embodiments of the mechanical support framework **200** (support framework **200**), array of cultivation pouches **310**, array of LEDs **410**, air and gas distribution unit **510**, and computer **520**.

[0045] Now referring to FIG. **2**, a perspective view of support framework **200** of the green microalga (*H. pluvialis*) farming assembly **100** in accordance with an exemplary embodiment of the present invention is illustrated. Support framework **200** functions as a skeletal support of farming assembly **100**. It houses, supports, and protects array of cultivation pouches **310**, array of LEDs **410**, air and gas distribution unit **510**, and computer **520**. Referring also to FIG. **3**, in various embodiments of the present invention, support framework **200** is a cuboid—a 3D shape having six faces and each face is a rectangle, also known as rectangular box shape. Support framework **200** further includes a main housing subunit **210** including a top rectangular frame **201**, a bottom rectangular frame **202**, and columns **203**—all are secured together by hinges **205** and screws **206**. At the four corners or vertices of bottom rectangular frame **202**, trolley or caster wheels **204** are attached. Support framework **200** also includes a LED support subunit **220** which is rectangular in shape and has the same dimension as support framework **200**. LED support subunit **220** includes a top bar **211**, a bottom bar **212**, outer side bars **213-214**, inner side bars **215-216**, and girts **217-218**. A pair of sliding wheels **219** is attached to bottom bar **212**. Pair of sliding wheels **219** helps LED support subunit **210** sliding in and out of main housing subunit **210** via bottom slider tracks **222**. Support framework **200** also includes top slider tracks **221** and bottom slider tracks **222** for receiving LED support subunit **220**. Top slider tracks **221** are parallel to one another and affixed to top rectangular frame **201**. Similarly, bottom slider tracks **222** are parallel to one another and secured to bottom rectangular frame **202**. As shown in FIG. **2**-FIG. **5** of the present invention, there are five top slider tracks **221** for coupling array of LED lights **310**. Intervening between top slider tracks **221** are pouch supporting bars **223**. Each pouch supporting bar **223** includes a series of hangers **225** and clip lines **224**. Clip line **224** seals off each cultivation pouch and is hung on to two adjacent hangers **225**.

[0046] Continuing with FIG. **3**, a perspective the structure and arrangement **300** of cultivation chambers (pouch) **310** of farming assembly **200** in accordance with an exemplary embodiment of the present invention is also illustrated. Array of cultivation pouches **310** contains a plurality of independent cultivation pouches **600**. Each cultivation pouch **600** is made of a soft durable and transparent medical grade plastic material such as polyethylene or linear low-density polyethylene (LLDPE). Each pouch supporting bar **223** hangs six single cultivation pouches **600** by means of hangers **225** and clip lines **224**. Each pouch supporting bar **223** is placed between two top slider tracks **221** so that LEDs **411** illuminate array of cultivation pouches **310**. As shown in FIG. **2**-FIG. **3**, there are four pouch supporting bars **223** connected to top rectangular frame **201**. For the description of an individual cultivation pouch **600** please also refer to FIG. **6**.

[0047] Next referring to FIG. **4**, a perspective diagram **400** showing an arrangement of array of LEDs **410** of the green microalga (*H. pluvialis*) farming assembly in accordance with an exemplary embodiment of the present invention is illustrated. Array of LEDs **410** includes a power supply and regulator **412** having a positive power supply output **413** and a negative power supply output **414**. Inner side bars **215-216**, top rectangular frame **201**, and columns **203** are hollow and contains electrical wiring bundle **420** that includes a positive electrical wire **421**, a negative electrical wire **422**, and a ground wire **423**. All electrical wires **421-423** are electrically coupled to male connectors **418** and female connectors **419** at both ends. Male connectors **418** and female connectors **419** are two pin clear insulating sleeve wire tube made of polyvinyl chloride (PVC). Negative power supply output **422** is electrically coupled to negative electrical wire **422**, positive power supply output **403** is to positive electrical wire **423**. Array of LEDs **410** is connected parallel to one another and adjacent to array of cultivation pouches **310**. As such, each LED support subunit **220** contains an array of LED lights **410**. Inner side bars **215-216** are hollow that contains electrical wiring bundle **420** connected to a series of electrical connectors **415** and **416**. Each LED light **411** is electrically connected to a pair of electrical connectors **415-416** respectively. Within the scope of the present invention, each array of LEDs **410** contains a plurality of individual LED **411**, each is selected at frequencies (colors) suitable for the growth of green microalgae (*H. pluvialis*). In some particular embodiments, LED light **411** is selected for the growth of keto-carotenoids at wavelengths of 442 nm to 472 nm, 24V with intensity 30 W to 130 W capable to varying the light intensity from 40 μmol photons/m2/s to 300 μmol photons/m2/s.

[0048] Next referring to FIG. **5**, a perspective diagram **500** of air and gas supply and distribution unit **510** (air and gas unit **510**) of farming assembly **100** in accordance with an exemplary embodiment of the present invention is illustrated. Air and gas **510** includes a carbon dioxide (CO.sub.2) gas source **511**, a gas pump **512**, and an air pump **513**. CO.sub.2 gas is administered to each cultivation pouch **600** in array of cultivation pouches **310** by a network of fluid tubes **519** which is connected to a first valve **513** and air pump **512**. In various aspects of the present invention, CO.sub.2 gas source **511** can be obtained from direct air capture technology, biomass carbon removal storage (BiCRS), or from direct air capture technology.

[0049] Air pump **512** is designed to pump carbon dioxide (CO.sub.2) to each cultivation pouch **600**

by a fixed dose controlled by first valve **513**. First valve **513** is a fluid metering device capable of delivering exact amount (dose) of CO.sub.2 to each cultivation pouch **600**. A gas meter **514** is used to monitor the exact dose of CO.sub.2 delivered to each cultivation pouch **600**. In some other embodiments of the present invention, gas pump **512** and first valve **513** are combined into one booster pump. An air pump **515** delivers an exact amount of oxygen to each cultivation pouch **311** by network of fluid tubes **513** that are connected to a second valve **516** and an air meter **517**. Again, the exact amount of oxygen (dose) can be controlled by second valve **516** and air meter **517**. Similarly, air pump **515** and second valve **517** can be combined into one booster pump. Booster pumps are pump that receives an input gas (either CO.sub.2 or oxygen) and delivers an exact amount of air or gas at a given pressure. Network of fluid tubes **519** is coupled to and supported by support frame **200**. Network of fluid tubes **519** leans against outer side bars **213** and bent along the length of top rectangular frame **201** to reach every single array of cultivation pouch **310**. The operations of gas pump **512**, first valve **513**, air pump **515**, and second valve **517** are controlled by computer **520**. In many embodiments of the present invention, gas pump **512** and air pump **513** are electrical solenoid valves which can be controlled by computer **520**.
[0050] Continuing with FIG. **5**, diagram **500** also illustrates monitoring and controlling unit **520** (computer **520**). Computer **520** includes a series of light sensors **526** located along the length of top rectangular frame **201** near each array of LED lights **310** and a series of temperature sensor and/or other sensors **527** (sensors **527**) submerged inside each cultivation pouch **311**. A computer **525** communicates to control signal generator **412** and light sensors **526**, temperature and/or other sensors **527** via either electrical wires or wireless channels **599**. In many embodiments of the present invention, wireless channels are wireless short-range communication channels, and long range wireless communication channels. Wireless short range communication channels include ZigBee™/IEEE 802.15.4, Bluetooth™, Z-wave, NFC, Wi-fi/802.11, cellular (e.g., GSM, GPRS, WCDMA, HSPA, and LTE, etc.), IEEE 802.15.4, IEEE 802.22, ISA100a, wireless USB, and Infrared (IR), LoRa devices, etc. Medium range wireless communication channels in this embodiment of communication link **599** include Wi-fi and Hotspot. Light sensors **526** are either diffuse reflective photoelectric type or through beam type (E-MSMT81P-2M). Temperature and/or other sensors **527** include pressure sensors, temperature sensors, and CO.sub.2 gas sensors. Pressure sensors are resistive force sensors. CO.sub.2 gas sensors are photoacoustic sensors. Temperature sensors are digital sensors that use diodes. These sensors—pressure, CO.sub.2 gas, and temperature—are available in the market and need not be discussed in details here.
[0051] Now referring to FIG. **6**, a 2D perspective diagram representing cultivation pouch **600** in accordance with an exemplary embodiment of the present invention is illustrated. As alluded above, each cultivation pouch **600** has a top portion **601**, a breather hole **602**, a body portion **603**, a bottom portion **604**. Bottom portion **604** has a taper shape while top portion **601** and body portion **603** have an even columnar shape. Top portion **601** is sealed by clip line **224** and suspended by hangers **225** to pouch supporting bar **223**. This way, fluid tubes **519** with a stone head **518** with sufficient pressure can create a vortex of an air circulation **621** from bottom portion **604** upward, creating a perfect habitat for green microalgae (*H. pluvialis*) **611** to grow. Temperature and/or other sensors **527** is submerged in cultivation pouch **600**. Cultivation pouch **600** contains a subtrate **605** conducive to the proliferation and accumulation of green microalgae (*H. pluvialis*) **611**. In many preferred embodiments of the present invention, substrate **605** is an Optimal Haematococcus Medium (OHM) that includes the following ingredients per liter: 0.41 g KNO.sub.3; 0.03 g Na.sub.2HPO.sub.4; 0.246 g MgSO.sub.4.Math.7H.sub.2O; 0.11 g CaCl.sub.2.Math.2H.sub.2O; 2.62 mg FeC.sub.6H.sub.5O.sub.7; 0.011 mg CoCl.sub.2; 0.012 mg CuSO.sub.4; 0.075 mg Cr.sub.2O.sub.3; 0.98 mg MnCl.sub.2; 0.12 mg Na.sub.2MoO.sub.4; 0.005 mg SeO.sub.2; 25 µg biotin; 17.5 µg thiamine; and 15 µg vitamin B.sub.12.
[0052] In operation, after farming assembly **100** is constructed as described above in FIG. **1** to FIG. **6**, a selected *H. pluvialis* strain is transferred from agar plates to OHM **605** inside each cultivation

pouch **600**. Array of LEDs **310** are switched on to a preselected intensities. These intensities will change as green microalgae (*H. pluvialis*) proliferates CO.sub.2 gas and air are pumped into each cultivation pouch **600** at preselected dose and pressure by means of air and gas unit **510** via respective gas valves **513** and air valves **516**. Computer **520** observes these preselect conditions and makes adjustments to obtain optimal living conditions for each colony of green microalgae (*H. pluvialis*) **611**. In many embodiments of the present invention, operation information from each cultivation pouch **600** in array of cultivation pouches **310** are networked and shared with other farming assemblies **100** in different geographical locations. This network arrangement achieves efficiency, useful information that reduces errors, infections, and achieves industrial production scale. The detailed methods of construction, culturing, harvesting with all preselected conditions are described in FIG. **7** to FIG. **14**.

[0053] Next referring to FIG. **7**, a flow chart of a method **700** for constructing a farming assembly that supports the farming and harvesting of green microalgae (*H. pluvialis*) in accordance with an aspect of the present invention is illustrated. Method **700** further includes a process **710** for constructing a single modular farming assembly **100** as described above and a process **711** for networking a plurality of culture assemblies located in different geographical locations. Modularity of farming assembly **100** enables a plurality of farming assemblies **100** to combine into a larger farming assembly at higher green microalgae output production.

[0054] At step **701**, a mechanical support framework is constructed. Step **701** is realized by support framework **200** as described in FIG. **2** and FIG. **3**. Supporting framework **200** further includes a main housing subunit **210** and a LED support subunit **220**.

[0055] Next, at step **702**, an array of cultivation pouches or chambers is constructed and coupled to the mechanical support framework. Step **702** is realized by array of cultivation pouches **310** as described in FIG. **2** and FIG. **3** above. Array of cultivation pouches **310** further contains a plurality of cultivation pouches **600** arranged next to one another. Array of cultivation pouches **310** is supported by pouch supporting bar **223** of support frame **200**. Each pouch supporting bar **223** contains row of six cultivation pouches **600**. Pouch supporting bars **223** are arranged parallel to each other and interspersed by top sliding tracks **221** also arranged in parallel to one another. It can be seen that cultivation pouches **600** is arranged in rows and columns as shown in FIG. **4**. The structure and content of each cultivation pouch **600** is described above in FIG. **6**.

[0056] At step **703**, an array of LEDs and electrical wiring (array of LEDs) is formed and mechanically coupled to the support framework. Step **703** is realized by array of LEDs and electrical wiring unit (array of LEDs) **410**. Array of LEDs **410** is coupled to top slider bars **221** of support frame **200**. Array of LEDs **410** is arranged in between array of cultivation pouches **310** so that LED lights can reach green microalgae (*H. pluvialis*) **611** inside each cultivation pouch **600**. Please refer to FIG. **4**-FIG. **5** for more detailed realization of step **703**.

[0057] At step **704**, an air and gas supply and distribution unit (air and gas distribution) is provided and coupled to the support framework. Step **704** is realized by air and gas distribution unit **510** (air and gas unit **510**) as described in FIG. **4** and FIG. **5** above. The object of step **704** is to regulate the exact amount of CO.sub.2 and air to each cultivation pouch **600**. Air and gas unit **510** of step **704** is coupled to and supported by support frame **200**.

[0058] At step **705**, a monitoring and controlling unit (computer) is provided and coupled to the support framework. Step **705** is realized by monitoring and controlling unit **520** (computer **520**) as described in FIG. **5**. Light sensors **526** and temperature and/or other sensors **527** send back signals reflecting habitat conditions of green microalgae (*H. pluvialis*) **611** inside each cultivation pouch **600** to computer **525**. In various aspects of the present invention, computer **525** is programmed to communicate to and control light sensors **526** and temperature and/or other sensors **527** via communication link **599**.

[0059] After step **705**, a single, modular, and independent farming assembly **100** is constructed that is designed to carry methods of the present invention for culturing green microalgae (*H. pluvialis*) and for harvesting at high efficiency.

[0060] Next at step **711**, a plurality of culture assembly built by steps **701-705** above are connected together by either networking and/or cloud networking. The detailed realization of step **706** is disclosed later in FIG. **12**-FIG. **14**. It is noted that different structures, configurations, steps can be used to realize steps **701-706** that produce either the same or obvious variations of farming assembly **100** described above in FIG. **1**-FIG. **6** are within the scope of the present invention.

[0061] Next, referring to FIG. **8**, a flow chart of a process **800** for cultivating green microalgae (*H. pluvialis*) and harvesting astaxanthin (keto-carotenoid) in according to an exemplary aspect of the present invention is illustrated and discussed.

[0062] After farming assembly **100** is constructed, process **800** of the present invention is performed to cultivate green microalgae (*H. pluvialis*) and harvest astaxanthin. Process **800** is comprised of a biomass growth phase **810** and an astaxanthin collection (harvest) phase **820**. In biomass growth phase **810**, green microalgae are allowed to proliferate in an acclimation phase, two continuous proliferation phases, and to encyst in an accumulation stage. In an acclimation phase, green microalgae (*H. pluvialis*) are allowed to acclimate to the OHM. In a first proliferation phase, the predetermined conditions are varied to cause cycle of unfavorable culture conditions and optimal culture are introduced to cause the green microalgae (*H. pluvialis*) to proliferate to 1.10.sup.5 cells/ml-3.10.sup.5 cells/ml. In a second proliferation phase, green microalgae (*H. pluvialis*) is subject to further proliferate to 10.sup.7 cells/ml-5.10.sup.8 cells/ml. In the accumulation stage, unfavorable conditions are introduced to change cellular morphologies from the green vegetative motile stage (macrozooids, microzooid, and pamella) to red vegetative non-motile haematocyst (aplanospores). In collection phase, the green microalgae (*H. pluvialis*) are removed and astaxanthin is harvested.

[0063] At step **811**, the biomass of the green microalgae (*Haematococcus Pluvialis*) is acclimated and proliferated. Step **811** is realized by first preparing an optimal *Haematococcus* medium (OHM). In many preferred aspects of the present invention, OHM is consisted of: 0.41 g KNO.sub.3; 0.03 g Na.sub.2HPO.sub.4; 0.246 g MgSO.sub.4.Math.7H.sub.2O; 0.11 g CaCl.sub.2.Math.2H.sub.2O; 2.62 mg FeC.sub.6H.sub.5O.sub.7; 0.011 mg CoCl.sub.2; 0.012 mg CuSO.sub.4; 0.075 mg Cr.sub.2O.sub.3; 0.98 mg MnCl.sub.2; 0.12 mg Na.sub.2MoO.sub.4; 0.005 mg SeO.sub.2; 25 µg biotin; 17.5 µg thiamine; and 15 µg vitamin B.sub.12. Furthermore, step **811** is realized by sub-steps:

[0064] Continued with step **811**, the acclimation and proliferation of green the microalgae (*H. pluvialis*) is introduced to a first predetermined vegetative condition. In the first predetermined conditions, the microalgae (*H. pluvialis*) is introduced to the OHM and allowed to acclimate to the medium. The light intensity slowly increases from 40 µmol photons/m2/s to 50 µmol photons/m.sup.2/s for 5 to 7 days. No carbon dioxide (CO.sub.2) is supplied and aeration (the supply of air) is supplied with a flow rate of 100 ml/min. Microalgae (*H. pluvialis*) proliferate slowly to 1.10.sup.5 cells/m. The proliferation rate in step **811** is 10%.

[0065] At step **812**, a first proliferation phase is performed. The first proliferation phase is specified by a second predetermined vegetative condition, in which: OHM increasing the temperature inside each cultivation pouch **600** at 20° C.-25° C. and continuously illuminating with a predetermined light intensity of 150 µmol photons/m.sup.2/s. In second predetermined vegetative conditions, CO.sub.2 gas is still not introduced and aeration is continued so that the microalgae (*H. pluvialis*) proliferate to 1.10.sup.5 cells/ml-3.10.sup.5 cells/ml. Inside each cultivation pouch **600**, the green microalgae (*H. pluvialis*) acts as phytoremediation agent which absorbs nutrients including pollutants such as carbon dioxide (CO.sub.2) and produces biomass with high value compounds such as astaxanthin.

[0066] At step **813**, a second proliferation phase is performed. Step **813** is realized by varying the second predetermined condition to a third predetermined vegetative condition. In step, CO.sub.2 is added for 14 days and other conditional factors stay the same. As discussed above, carbon dioxide

(CO.sub.2) source can be obtained from carbon dioxide source **511** such as direct air capture technology, biomass carbon removal storage (BiCRS), or from direct air capture technology. The green microalgae (*H. pluvialis*) proliferates to 10.sup.7 cells/ml-5.10.sup.8 cells/ml. More specifically, the CO.sub.2 gas is started with a flow rate from 1 mg/l/min to 5 mg/l/min, air is supplied at a flow rate of 2.5 liters/minute to 4 liters/minute to provide nutrition and proliferation for the green microalga (*H. pluvialis*). Air flows at the rate of 2.5 liters/minute to 4 liters/minute to stir the green microalgae (*H. pluvialis*) at the bottom of cultivation pouch **600**. As such, clogging of the green microalgae *H. pluvialis* is prevented. The biomass growth period ranges from 7 to 15 days.
[0067] At step **814**, the accumulation phase is begun. Step **814** is realized by
[0068] a fourth predetermined vegetative condition. In the fourth predetermined vegetative condition, adverse conditions are introduced to stress out the green microalgae (*H. pluvialis*) forcing them to produce more astaxanthin. The light intensity is increased to 300 μmol photons/m.sup.2/s-400 μmol photons/m.sup.2/s. Temperature are increased to 28° C.-30° C. Micronutrients such as sodium acetate (C.sub.2H.sub.3NaO.sub.2), sodium chloride (NaCl) are added. The ratio of mass/volume of micronutrients added to the vegetative environment ranges from 0.5 g/l to 2.5 g/l. At this stage, air is continuously supplied into cultivation pouch **600** to aid in the accumulation astaxanthin with a flow rate of 2.5 liters/minute to 4 liters/minute. After a period of astaxanthin accumulation of about 10 to 15 days, the density of green microalga (*H. pluvialis*) **611** accumulating astaxanthin in each cultivation pouch **600** decreases to 10.sup.6 cells/ml-5.10.sup.7 cells/ml. Please refer to FIG. **10**C for reference.
[0069] In summary, from steps **811-814**, green microalga (*H. pluvialis*) cells are subject to different predetermined vegetative conditions. When exposed to their favorable conditions, they grow into a characteristic green color with increase in biomass. However, when exposed to unfavorable environmental conditions in terms of nutrients, temperature, pH, light, etc., they change from green vegetative motile cells to red non-motile haemotocysts that accumulate astaxanthin.
[0070] Now, the harvest of astaxanthin phase **820** begins:
[0071] At step **821**, biomass of green microalga (*H. pluvialis*) is collected by dehydration (removal of water) method. In some aspects of the present invention, water is removed by centrifugation or vacuum filtration. According to a specific aspect, after the green microalga (*H. pluvialis*) cells change from red nucleus to completely red. At this time, the green microalga (*H. pluvialis*) cells have accumulated rich astaxanthin content. They proceeds to obtain biomass. Specifically, medium **605** from array of cultivation pouches **310** is continuously pumped into the centrifugation device at a rate of 15,000 rpm until it runs out. The algal biomass after centrifugation is prepared for the algae freeze-drying process.
[0072] At step **822**, the green microalgae biomass is freeze dried. Specifically, the green microalga (*H. pluvialis*) biomass after centrifugation is frozen at −20° C. for 48 hours. Then it is lyophilized at −5° C. and at 0.1 mBar pressure until the biomass is completely dried. Other drying methods include spray drying, or cryodesiccation.
[0073] At step **823**, astaxanthin-rich green microalga powder (*H. pluvialis*) is collected. Step **823** is realized by storing the freeze dried biomass it in vacuum bags, avoiding light exposure and storing at −20° C. According to one aspect of the present invention, the mass of astaxanthin-rich green microalga (*H. pluvialis*) powder obtained at least 2.5 g/l; and astaxanthin content reaches at least 5%.
[0074] Continuing with step **823**, green microalgae powder (*H. pluvialis*) rich in astaxanthin is extracted and astaxanthin is quantified. The extraction method is performed as follows: weigh the green microalga (*H. pluvialis*) powder into a tube, add 4M hydrochloric acid (HCl). The mixture is incubated at 70° C. for 5 minutes. The treated mixture is centrifuged at 12,000 rpm for 10 minutes at 4° C. The HCl acid solution is removed, washed twice with sterile distilled water. Then, the solution is centrifuged again to remove water. Green microalgae biomass is dissolved in 1 ml of

99% acetone solvent. The mixture is sonicated in a cold bath for 20 minutes, then centrifuged at 13,000 rpm/5 minutes at 4° C. The supernatant containing astaxanthin is collected. The extraction method with acetone solvent is repeated until the algae biomass completely lost its color. The final acetone extract is filtered through a 0.22 μM filter and prepared for HPLC-High Performance Liquid Chromatography analysis. Please refer to FIG. **11**-FIG. **12**.

[0075] The freeze-dried microalgal biomass sample is then extracted and determined astaxanthin content using HPLC high-performance liquid chromatography. In realization of step **823**, samples were sent for analysis at the Ho Chi Minh City Laboratory Analysis Service Center. The standard substance (all-trans-astaxanthin) was mixed into many different concentrations, then these concentrations were analyzed by HPLC to determine the peak areas. From the peak areas of the standard concentrations, a standard curve equation of the form y=ax+b was, in which y is the peak area, x is the astaxanthin content. The green microalgae (*H. pluvialis*) powder sample was extracted with the acetone solvent mentioned above and the extract was analyzed by HPLC. The peak area results and the standard curve were used in the equation above to calculate the astaxanthin content in the extract of the green microalgae (*H. pluvialis*) powder.

[0076] Now referring to FIG. **9**, a flow chart of a software program application **900** (process **900**) for farming and harvesting green microalgae (*H. pluvialis*) in a network of culture assemblies described in FIG. **1**-FIG. **6** in accordance with an exemplary aspect of the present invention is illustrated. Process **900** is stored in a non-transitory memory and executed by a central processing unit (CPU) (see FIG. **15** and FIG. **16**). The objective of process **900** is to: (a) controlling the cultivation and harvesting processes of green microalgae in a network of farming assemblies **100**, (b) sharing viable information to increase overall efficiency and growth rate, and (c) avoid errors and infections in farming assemblies **100**.

[0077] At step **901**, previous successful farming information is obtained. Step **901** is realized by light sensors **526** and temperature and other sensors **527** and computer **525** from each farming assembly **100** within the network (e.g. green algae farming network **1400**). Successful farming information and experience are stored in storage device accessible to all computers **525** within the farming network.

[0078] Next, at step **902**, previous failing farming information is also obtained and loaded into computers. Step **902** is realized by sensors light sensors **526**, temperature and other sensors **527**, and computer **525** from each farming assembly **100** within the network. As alluded before, every time any cultivation pouches **600** among array of cultivation pouches **310** within the network fails and/or encounters challenges, they are replaced by new cultivation pouch **600** and step **810** is performed again. These failed cultivation pouches **600** are analyzed and failure information is stored in computer **525** as useful guide for future development. The review of the successful and/or failed information serve as instruction information for future machine learning.

[0079] At step **903**, a biomass growth process described in step **810** is performed. After obtaining successful and failure information, array of cultivation pouches **310** are prepared for biomass growth as described in step **810**.

[0080] At step **904**, temperature, pressure, air content, CO.sub.2 content, and flow rate for each cultivation pouch **600** in array of cultivation pouches **310** are monitored and determined whether they follow the predetermined vegetative conditions. The predetermined vegetative conditions are disclosed in FIG. **8** above.

[0081] At step **905**, when at least one cultivation pouch is failed and predetermined vegetative conditions change leading to low biomass count, those cultivation pouches within array of cultivation pouches **310** are replaced by new ones and step **903** repeats.

[0082] At step **906**, when predetermined vegetative conditions are met and the green microalgae are ready for harvest, go to step **907**.

[0083] At step **907**, failure and successes as well as locations of each cultivation pouch **600** are statistically analyzed and stored in memory storage (see FIG. **15** and FIG. **16**). Statistical analyses including (but not limited to) means, standard deviations, analysis of variance (ANOVA), distribution functions for successes and failures are performed. These analytics and operation information help in the total production and overall efficiency of green microalgae farming.

[0084] At step **908**, the statistical analyses and operation information are shared among cultivation pouches **600** within the network (please refer to FIG. **14**-FIG. **16**). Finally, when green microalgae (*H. pluvialis*) are ready to harvest, step **820** described above is performed.

[0085] FIG. **10**A-FIG. **10**C illustrate step **906** for determining whether green microalgae **611** in each cultivation pouch **600** is ready for harvest. In other words, FIG. **10**A-FIG. **10**C illustrate the cellular morphologies of green microalgae (*H. pluvialis*) from green vegetative motile stage to red non-motile haematocyst stage. According to FIG. **10**A, in the proliferation stage with optimal conditions, the green microalga (*H. pluvialis*) **1001** cells are green vegetative motile stage and divide into many nearly circular cells. Normally after about 10 days of culture, according to FIG. **10**B, the green microalgae cell (*H. pluvialis*) **1011** encloses a red nucleus **1012** in the middle. At this time the green microalgae cell (*H. pluvialis*) is at the end of the growth phase. The nutrient medium begins to exhaust. Therefore, after the astaxanthin accumulation period, green microalgae (*H. pluvialis*) cells **1021** grow larger and reach red non-motile haematocyst stage as shown in FIG. **10**C. This is a sign of the end of the astaxanthin accumulation phase. The extraction of astaxanthin-rich in green microalgae (*H. pluvialis*) begins.

[0086] According to FIG. **11**, a HPLC high-performance liquid chromatography chart **1100** of an extract sample from green microalgae (*H. pluvialis*) powder is illustrated. Based on the HPLC high performance liquid chromatography chart **1100** and Table 1, the astaxanthin content in green microalgae (*H. pluvialis*) powder extract is calculated to be 5.09%.

TABLE-US-00001 TABLE 1 The Peak Information and Area of the Sample Extracted from Green Microalgae (*H. Pluvialis*) Powder with Astaxanthin Content of 5.09%. No. Peak Area Height Area (%) Height (%) 1 5.56 513690 35657 86.273 88.116 2 6.286 6719 415 1.128 1.025 3 7.006 48293 2788 8.111 6.891 4 7.537 26723 1606 4.488 3.968 Total 595424 40466 100 100

[0087] According to FIG. **12**, a HPLC high-performance liquid chromatography chart **1200** of an extract sample from green microalgae (*H. pluvialis*) powder is illustrated. Based on the HPLC high performance liquid chromatography chart **1100** and Table 2, the astaxanthin content in green microalgae (*H. pluvialis*) powder extract is calculated to be 7.92%.

TABLE-US-00002 TABLE 2 The Peak Information and Area of the Sample Extracted from Green Microalgae (*H. Pluvialis*) Powder with Astaxanthin Content of 7.92%. No. Peak Area Height Area (%) Height (%) 1 6.043 741448 53643 78.272 81.155 2 6.563 6079 422 0.642 0.639 3 7.173 63515 3860 6.705 5.84 4 7.657 136232 8174 14.382 12.366 Total 947275 66099 100 100

[0088] According to FIG. **13**, a HPLC high-performance liquid chromatography chart **1300** of an extract sample from green microalgae (*H. pluvialis*) powder is illustrated. Based on the HPLC high performance liquid chromatography chart **1100** and Table 1, the astaxanthin content in green microalgae (*H. pluvialis*) powder extract is calculated to be 10.09%.

TABLE-US-00003 TABLE 3 The Peak Information and Area of the Sample Extracted from Green Microalgae (*H. Pluvialis*) Powder with Astaxanthin Content of 10.09%. No. Peak Area Height Area (%) Height (%) 1 5.986 1069025 64890 81.662 83.788 2 6.45 2846 264 0.217 0.341 3 7.106 65001 3488 4.965 4.504 4 7.575 172218 8803 13.156 11.367 Total 1309090 77445 100 100

[0089] Thus, the high and stable astaxanthin content >5% has proven the effectiveness of the green microalga *H. pluvialis* farming method according to the invention.

[0090] The results of the astaxanthin content extracted from the green microalga (*H. pluvialis*) as described in FIG. **1** to FIG. **9** of the present invention is compared to other companies around the world including: Algalif (Iceland) and Algamo (Czech) company have a stable astaxanthin content at 5%, Astareal company (Japan) has a stable content of 5%. astaxanthin stabilized at 4%, NatAxtin company. The comparison is tabulated in Table 4.

TABLE-US-00004 TABLE 4 Comparison of the Astaxanthin Content Extracted from the Green microalga (*H. pluvialis*) between Companies in the world and the Present Invention. The Publication of the Concentration of Astaxanthin Extracted from Microalgae No. Company (*H. pluvialis*) 1 Algalif (Iceland) 5% 2 Algamo (Czech) 5% 3 Astareal (Japan) 4% 4 NatAxtin (Chile) 3%-4% 5 Algatechnologies (Israel) 2.5%-3% 6 Bluebiotech (Germany) 1.5-5% 7 YunNan (China) 1.5%-3% 8 Norland (China) 1.5%-3% 9 Cyanotech (USA) 1%-2% 10 Present Invention >5%

[0091] Now referring to FIG. **14**, a systematic diagram of a green microalgae (*H. pluvialis*) network **1400** (farming network **1400**) in accordance with an exemplary embodiment of the invention is illustrated. In many embodiments, algae farming network **1400** includes a first algae farm **1400-1**, a second algae farm **1400-2**, . . . and an J.sup.th algae farm **1400-**J, where J is a non-zero positive number. Each algae farm **1400-1**, **1400-2**, . . . **1400-**J further includes a plurality of farming assemblies similar to farming assembly **100** as described in FIG. **1** or one constructed by method **700** described in FIG. **7**. Specifically, first algae farm **1400-1** is located in distant geographical area such as Binh Duong province in VN and comprised a plurality of culture assemblies **100-11**, **100-12**, . . . , **100-1**N, each is connected to one another by a local network **1401-1**. Second algae farm **1400-2** is located in a different geographical area such as Lam Dong province in VN and comprised a plurality of culture assemblies **100-21**, **100-12**, . . . , **100-2**M, each is connected to one another by a local network **1401-2**. J.sup.th farm **1400-**J is located in a distant geographical area such as Vinh Phuc province in VN and comprised a plurality of culture assemblies **100-**J**1**, **100-**J**2**, . . . , **100-**JM, each is connected to one another by a local network **1401-**J. Alternatively, first algae farm **1400-1**, second algae farm **1400-2**, . . . and J.sup.th algae farm **1400-**J are located near one another in one slot of land.

[0092] Continuing with FIG. **14**, first algae farm **1400-1**, second algae farm **1400-2**, and J.sup.th algae farm **1400-**J are connected together by a network **1420** via communication channels **1499**. Communication channels **1499** are wireless short-range communication channels, and long range wireless communication channels. Wireless short range communication channels include ZigBee™M/IEEE 802.15.4, Bluetooth™, Z-wave, NFC, Wi-fi/802.11, cellular (e.g., GSM, GPRS, WCDMA, HSPA, and LTE, etc.), IEEE 802.15.4, IEEE 802.22, ISA100a, wireless USB, and Infrared (IR), LoRa devices, etc. when these algae farms are located near one another. Medium range wireless communication channels in this embodiment of communication link **1499** include Wi-fi and Hotspot. Long range wireless communication channels include UHF/VHF radio frequencies when algae these farms are located far from one another. Farm network **1400** also includes a plurality of local hubs or gateways **1410-1**, **1410-2**, . . . **1410-**I that are also connected to network **1420** and to first algae farm **1400-1**, second algae farm **1400-2**, . . . and J.sup.th algae farm **1400-**J. End-users **1411** such as farmers, owners, scientists, researchers can access to view and control each cultivation pouch **600** in first algae farm **1400-1**, second algae farm **1400-2**, . . . , and J.sup.th algae farm **1400-**J via a software application program described in FIG. **9** rendered by a graphic user interface (GUI) (please see FIG. **16**). A server computer **1540** is also connected to network **1420** via wireless communication channel **1499** to manage, store, and operate farm network **1400**. Server computer **1540** may be a personal desktop computer, a laptop, a smart phone, or a tablet.

[0093] Next referring to FIG. **15**, a schematic diagram of a server system **1500** that manage the farm network in accordance with an exemplary embodiment is illustrated. Server system **1500** includes a server computer **1540** electrically coupled to local gateways **1410-1** and to other distant gateways **1410-1**, **1410-1** to **1410-**N via a communication channel **1499** to network **1420** Server computer **1540** includes, but not limited to, input/output interface **1534**, a storage device **1520**, a central processing unit (CPU) **1530**, a ROM/RAM for temporary memories **2250**, a display unit **1537**, a power supply unit **1531**, an internet interface **1532**, an audio/video board **1535**, a keyboard **1536**, and a transceiver **1539**, all electrically coupled to one another via electrical connections **1599**. These units are well-known computer components in the arts and need not be described in details here. It is noted that, non-limiting examples of network **1420** include the internet, cloud computing, Software as a service (Saas), Platform as a service (PaaS), Infrastructure as a service (IaaS), or permanent storage such as optical memory (CD, DVD, HD-DVD, Blue-Ray Discs), semiconductor memory (e.g., RAM, EPROM, EEPROM), and/or magnetic memory (hard-disk drive, floppy-disk drive, tape drive, MRAM) among others.

[0094] Storage device **1520** stores a OS/BIOS **1521** and a memory banks **1522**, and a program green microalgae culture system controller **1510** (system controller **1510**) In various embodiments of the present invention, server computer **1540** is a printed circuit board (PCB) with electrical connections **1599** are conducting wires such as copper, aluminum, gold, etc. In operation, server computer **1540** operates like a regular computer except that it is load with system controller **1510**. System controller **1510** is executed by CPU **1530**. Transceiver **1539** sends and receives information and instructions to each cultivation pouch **600** in array of cultivation pouches **310** in each algae farms **1400-1**, **1400-2**, . . . , and **1400**-J. System controller **1510** is a software program described by FIG. **9** of the present invention.

[0095] Referring again to FIG. **15**, system controller **1510** further includes a feedback data analyzer **1511**, a controller module **1512**, and a GUI module **1513**. Feedback data analyzer **1511** receives green microalgae (*H. pluvialis*) information from each cultivation pouch **600** in array of cultivation pouches **310** in each algae farms **1400-1**, **1400-2**, . . . , and **1400**-J and performs analytics on this information. Controller modules **1512** sends appropriate instructions to perform and maintain "homeostasis" of process **800**. GUI module **1513** provides a display interface (see FIG. **16**) so that end-users **1411** can view and control farm network **1400**.

[0096] Referring to FIG. **16**, a Graphical User Interface (GUI) displayed on the server computer for monitoring and controlling farm server **1400** in accordance with an exemplary aspect of the present invention is illustrated. Therein, GUI **1600** includes a number of function buttons from function button **1661** to function button **1670** arranged around a control and monitor area **1601**. Control and monitor area **1601** provides data information representation of sensors **526** and **527** (see FIG. **5**) in each farming assembly **100** and networked culture assemblies **100-11**. **100-12** . . . **100-1**N in first algae farm **1400-1**; **100-21**, **100-22**, . . . **100-2**M in second algae farm **1400-2**; and **100**-J**1**, **100**-J**2**, . . . **100**-JQ in J.sup.th algae farm. Specifically, some specific function buttons include but are not limited to function buttons: FILE button **1661** (file button); DESIGN button **1662** (table setup button); EDIT button **1663** (adjustment button); SINGLE VIEW button **1664** (showing a single farming assembly **100** as currently presented in FIG. **16**); GROUP VIEW button **1665** (multiple culture farms **100** views); ANALYTICS button **1666** (data analysis from culture assemblies **100-11**, **100-12**, . . . **100-1**N in first algae farm **1400-1**; **100-21**, **100-22**, . . . **100-2**M in second algae farm **1400-2**; and **100**-J**1**, **100**-J**2**, . . . , **100**-JQ in J.sup.th algae farm 'farm network **1400**'); IMPORT DATA button **1667** (button to import data from farm network **1400**); EXPORT DATA button **1668** (data export button to and from farm network **1400**); SETTING button **1669** (GUI interface adjustment button); HOME button **1670** (button to return to default display).

[0097] Continuing with FIG. **16**, FILE button **1661** (file button) is a drop-down menu that includes many sub-functions such as opening previous file, saving file; print file; closes the current file or window and the current information of the open file. DESIGN button **1662** (presentation setting button) helps select the presentation of data obtained from the sensor card in the form of tables and graphs, such as table shown in control and monitor area **1601**. EDIT button **1663** allows changes to the data information of the file being displayed in control and monitor area **1601**. SINGLE VIEW button **1664** and GROUP VIEW button **1665** displays data information from one or multiple farming assembly **100** of farm network **1400**. ANALYTICS button **1666** performs analysis such as statistical analysis (e.g., analysis of variance 'ANOVA'), calculations. DESIGN button **1662** creates tables/charts for displaying essential information after analyzing data collected from farm network **1400**.

[0098] Still referring to FIG. **16**, IMPORT DATA button **1667** allows end-users **1411** to import data from a data storage device **1520** network hard drives or from cloud databanks, into the user interface **1600** to use function buttons from the user interface **1610** such as data analysis via the ANALYTICS button **1666**. EXPORT DATA button **1668** export of data files containing farming data to and from other culture assemblies **100** in farm network **1400**. SETTING button **1669** changes the default predetermined functions of user interface **1600** or the control and monitor area **1601**. HOME button **1670** when executed returns to the screen display showing control and monitor area **1601**.

[0099] Although the implementation options of the present invention are disclosed through the detailed description of the invention above, however, it should be understood that the invention is by no means limited to these implementation options. Experts in the same technical field admit that many other similar changes and arrangements could be made. Therefore, the scope of the invention is clearly defined to include all similar changes and arrangements within the scope of the following attached claims.

EXPLANATION OF REFERENCE NUMERALS

[0100] **100** green microalgae (*H. Pluvialis*) farming assembly (assembly) [0101] **100-11** farming assembly in first algae farm [0102] **100-21** farming assembly in second algae farm [0103] **100**-J**1** farming assembly in J.sup.th algae farm [0104] **200** mechanical support framework (framework) [0105] **201** top rectangular frame [0106] **202** bottom rectangular frame [0107] **203** columns [0108] **204** caster or trolley wheels [0109] **205** hinges [0110] **206** screws [0111] **210** main housing unit [0112] **211** top bar [0113] **212** bottom bar [0114] **213** left outer side bar [0115] **214** right outer side bar [0116] **215** left inner side bar [0117] **216** right inner side bar [0118] **217** girt (reinforcement bar) [0119] **218** girt (reinforcement bar) [0120] **219** sliding wheels [0121] **220** LED lights support framework [0122] **221** top slider tracks [0123] **222** bottom slider tracks [0124] **223** pouch supporting bars [0125] **224** clip lines [0126] **225** hangers [0127] **310** array of cultivation pouches [0128] **410** LED and electrical wiring unit [0129] **411** LED light [0130] **412** power supply and/or regulator [0131] **413** positive power supply output [0132] **414** negative power supply output [0133] **418** male 2 pin insulated cap connectors [0134] **419** female 2 pin insulated cap connectors [0135] **421** positive power wire [0136] **422** negative power wire [0137] **423** ground wire [0138] **510** air and gas supply and distribution unit [0139] **511** CO.sub.2 gas source [0140] **512** gas pump [0141] **513** gas valve [0142] **514** gas meter [0143] **515** air pump (aerator or mixer) [0144] **516** air valve [0145] **517** air meter [0146] **518** air stone [0147] **519** fluid tubes [0148] **525** computer [0149] **526** light sensors [0150] **527** temperature and other sensors such as pressure [0151] **520** monitoring and controlling unit [0152] **529** communication link [0153] **600** cultivation pouch [0154] **601** top portion of cultivation pouch [0155] **602** breather hole [0156] **603** body portion [0157] **604** bottom portion [0158] **605** culturing medium [0159] **611** green microalgae colony inside each cultivation pouch [0160] **621** air flow [0161] **1400** green algae farming network [0162] **1400-1** first algae farm [0163] **1400-2** second algae farm [0164] **1400-**J J.sup.th algae farm [0165] **1401-1** first local network [0166] **1401-2** second local network [0167] **1401-**J J.sup.th local network [0168] **1410-1** first gateway [0169] **1410-2** second gateway [0170] **1410-**J J.sup.th gateway [0171] **1411** end-users [0172] **1499** wireless communication channels (links) [0173] **1500** server gateway system [0174] **1510** green microalgae culture system controller [0175] **1511** feedback & data analyzer module [0176] **1512** controller module [0177] **1513** GUI unit [0178] **1520** storage device [0179] **1521** OS/BIOS [0180] **1522** non-transitory memories [0181] **1530** CPU [0182] **1531** power supply [0183] **1532** internet interface [0184] **1533** ROM/RAM [0185] **1534** I/O interface [0186] **1535** audio/video [0187] **1536** keyboard [0188] **1537** display [0189] **1538** I/O driver [0190] **1539** transceiver [0191] **1540** server computer [0192] **1600** GUI [0193] **1601** display area [0194] **1602** cursor or pointing device [0195] **1603** single algae farming information display [0196] **1661** file command [0197] **1662** design command [0198] **1663** edit command [0199] **1664** single view [0200] **1665** group view [0201] **1666** analytics function [0202] **1667** import data [0203] **1668** export data [0204] **1669** setting command [0205] **1670** home command

Claims

1. A green microalga farming assembly, comprising: a mechanical support framework; a plurality of cultivation pouches, mechanically coupled to said mechanical support framework, having at least two arrays of cultivation pouches where green microalgae (*H. pluvialis*) are cultured; a plurality of light emitting diodes (LEDs), mechanically coupled and supported by said mechanical support framework and arranged in between said at least two arrays of cultivation pouches, operable to provide and vary light intensities to each of said at least two arrays of cultivation pouches according to predetermined vegetative conditions; and an air and gas supply and distribution unit, mechanically coupled and supported by said mechanical support framework, operable to supply and vary amounts of air and carbon dioxide (CO.sub.2) to said at least two arrays of cultivation pouches according to said predetermined vegetative conditions.
2. The microalgae farming assembly of claim 1, further comprising a monitoring and controlling computer, mechanically coupled and supported by said mechanical support framework, operable to maintaining said predetermined vegetative conditions.
3. The microalgae farming assembly of claim 1, wherein said monitoring and controlling computer further comprises a plurality of temperature sensors, pressure sensors, light intensities sensors, and CO.sub.2 sensors coupled to and supported by said mechanical support framework, operable to measure temperature, pressure, light intensities, and CO.sub.2 gas in said at least two arrays of cultivation pouches.
4. The microalgae farming assembly of claim 1, wherein said mechanical support framework further comprises: a main housing subunit having a cuboid shape; a cultivation pouch supporting frame mechanically coupled to said main housing unit; and LED supporting frames mechanically coupled to said main housing subunit.
5. The microalgae farming assembly of claim 4, wherein said main housing subunit further comprises: a top rectangular frame; a bottom rectangular frame; and a plurality of columns connected to said top rectangular frame and bottom rectangular frame so as to form said cuboid shape.
6. The microalgae farming assembly of claim 5, wherein said cultivation pouch support frame further comprises a pouch hanging bar connected to said top rectangular frame for suspending said plurality of cultivation pouches.
7. The microalgae farming assembly of claim 6, wherein said LED supporting frames further comprises: a top bar; a bottom bar; and at least two side bars connected to said top bar and said second bar, operable to electrically coupled to said plurality of LEDs; wherein said plurality of LEDs within said LED light support frame is arranged in between any two of said arrays of cultivation pouches.
8. The microalgae farming assembly of claim 1, wherein said air and gas supply and distribution unit further comprises: an carbon dioxide (CO.sub.2) source; an gas valve; a gas meter; and a plurality of fluid tubes connected to said carbon dioxide source, said gas valve, and said gas meter to provide CO.sub.2 gas to each of said at least two arrays of cultivation pouches.
9. The microalgae farming assembly of claim 8, wherein said air and gas supply and distribution unit further comprises: an air pump; an air valve; an air meter; and said plurality of fluid tubes connected to said air pump, said air valve, and said air meter provide CO.sub.2 gas to each of said at least two arrays of cultivation pouches.
10. The microalgae farming assembly of claim 1, wherein each of said cultivation pouches is made of a transparent durable material capable of receiving lights from said plurality of LEDs.
11. The microalgae farming assembly of claim 10, wherein each of said at least two arrays of cultivation pouches comprises an optimal *Haematococcus* Medium (OHM) consisted of 0.41 g KNO.sub.3; 0.03 g Na.sub.2HPO.sub.4; 0.246 g MgSO.sub.4.Math.7H.sub.2O; 0.11 g

CaCl.sub.2.Math.2H.sub.2O; 2.62 mg FeC.sub.6H.sub.5O.sub.7; 0.011 mg CoCl.sub.2; 0.012 mg CuSO.sub.4; 0.075 mg Cr.sub.2O.sub.3; 0.98 mg MnCl.sub.2; 0.12 mg Na.sub.2MoO.sub.4; 0.005 mg SeO.sub.2; 25 μg biotin; 17.5 μg thiamine; and 15 μg vitamin B.sub.12 per one liter.

12. The microalgae farming assembly of claim 11, wherein said monitoring and controlling computer unit is loaded with a software application program stored in a non-transitory memory when executed by a central processing unit (CPU), said software application program is operable to perform the following steps: (a) a biomass growth phase in which green microalgae in said at least two arrays of cultivation pouches are subject to said predetermined vegetative conditions in which said light intensities increase to 100 μmol photons/m.sup.2/s, air is supplied, and temperature is maintained at 25° C.-28° C.; and (b) an accumulation phase in which said temperature is changed to 28° C.-30° C. and light intensities increases to 300-400 μmol photons/m.sup.2/s.

13. The microalgae farming assembly of claim 12, further comprising a step of harvesting astaxanthin by by removing water and freeze drying said green microalgae (*H. pluvialis*).

14. A method of farming and harvesting green microalgae (*H. pluvialis*), comprising: (a) constructing a framework that supports a cultivation for said green microalgae (*H. pluvialis*); (b) a biomass growth phase in which green microalgae (*H. pluvialis*) are subject to predetermined vegetative conditions in which said light intensities increase to 100 μmol photons/m.sup.2/s, air is supplied, and temperature is maintained at 25° C.-28° C.; and (c) an accumulation phase in which said temperature is changed to 28° C.-30° C. and said light intensities increases to 300-400 μmol photons/m.sup.2/s.

15. The method of claim 14 further comprising a step of harvesting astaxanthin from said green microalgae (*H. pluvialis*) by removing water and freeze drying said green microalgae (*H. pluvialis*).

16. The method of claim 14, wherein said predetermined vegetative conditions further comprises cultivating said green microalgae (*H. pluvialis*) in an Optimal *Haematococcus* Medium (OHM) that consisted of 0.41 g KNO.sub.3; 0.03 g Na.sub.2HPO.sub.4; 0.246 g MgSO.sub.4.Math.7H.sub.2O; 0.11 g CaCl.sub.2.Math.2H.sub.2O; 2.62 mg FeC.sub.6H.sub.5O.sub.7; 0.011 mg CoCl.sub.2; 0.012 mg CuSO.sub.4; 0.075 mg Cr.sub.2O.sub.3; 0.98 mg MnCl.sub.2; 0.12 mg Na.sub.2MoO.sub.4; 0.005 mg SeO.sub.2; 25 μg biotin; 17.5 μg thiamine; and 15 μg vitamin B.sub.12 per one liter.

17. The method of claim 16, wherein said constructing a green microalgae farming assembly further comprises: constructing a mechanical support framework; constructing and coupling at least two arrays of cultivation pouches, to said mechanical support framework, wherein said at least two arrays of cultivation pouches contain said OHM; constructing and coupling at least two arrays of light emitting diodes (LEDs), to said mechanical support framework and arranging said at least two arrays of LEDs in between said at least two of arrays of said cultivation pouches, wherein said LEDs are operable to provide said predetermined light intensities to each of said at least two of arrays of said cultivation pouches; and constructing and coupling an air and gas distribution unit to said mechanical support framework, wherein said air and gas distribution unit is operable to supply predetermined amounts of air and carbon dioxide (CO.sub.2) to each of said at least two of arrays of said cultivation pouches.

18. The method of claim 14, wherein said step (a) of constructing a framework further comprises: constructing and coupling a monitoring and controlling computer to said mechanical support framework; wherein said monitoring and controlling computer is operable to maintaining said predetermined light intensities and said predetermined amounts of air and carbon dioxide (CO.sub.2) to each of said at least two of arrays of said cultivation pouches.

19. The method of claim 18, wherein said step of constructing and coupling a monitoring and controlling computer further comprises a plurality of temperature sensors, pressure sensors, light intensities sensors, and gas sensors mechanically coupled to said mechanical support framework, operable to measure temperature, pressure, CO.sub.2 and air, and light intensities in each of said at least two of arrays of said cultivation pouches.

20. Method of farming and harvesting green microalgae (*H. pluvialis*) of claim 14 further comprising: networking said plurality of said green microalgae farming assemblies by a cloud network and a server computer, wherein said server computer is loaded with a software application stored in a non-transitory memory, wherein when executed by a central processing unit (CPU) of said server computer, said software application is operable to perform the following steps: monitoring said predetermined vegetative conditions in each of said at least two of arrays of said cultivation pouches; and if said said predetermined vegetative conditions are different from said light intensities and amounts of air and carbon dioxide (CO.sub.2), adjusting said temperature, said pressure, said CO.sub.2 and air, and said light intensities to said predetermined light intensities and said amounts of air and carbon dioxide (CO.sub.2).